

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0518

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

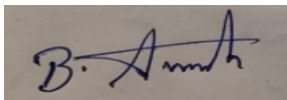
Assessment Tool Weightage: 40%

Code of Conduct:

I Avinash B (192511193) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in blue ink, appearing to read 'B. Avinash', is written on a light-colored rectangular background.

Question 3 – Optical Flow Reliability vs Noise Sensitivity

The given comparison contains two optical flow methods. Method A produces stable vectors but misses small motions. Method B is able to detect fine motion, but its vectors are noisy. Therefore, the main trade-off is between stability and sensitivity to small movements.

Method A is useful when the system needs smooth and reliable motion vectors. Stable vectors reduce fluctuations in the estimated motion and make tracking more consistent. Its major limitation is that small or subtle movements may not be detected. This can be a problem in applications where small object movements are important.

Method B has better sensitivity to fine motion. This means it can identify small movements that Method A may miss. However, noisy vectors can create unstable tracking, false motion indications, and extra processing requirements for filtering or smoothing.

For dynamic scenes, the best method depends on the application. If the main requirement is stable long-term motion tracking, Method A is more suitable. If the application focuses on detecting very small movements and can tolerate or reduce noise using filtering techniques, Method B is preferable.

Conclusion

Method A is the safer general choice for reliable motion tracking because stable vectors provide consistent results. Method B should be selected when fine-motion detection is more important than vector stability. The assessment specifically presents this stability-versus-fine-motion trade-off.

□filecite□turn1file0□L37-L42□

Question 27 – Video Processing Pipeline Comparison

The three pipelines have different accuracy and speed characteristics. Pipeline A provides 88% accuracy and is fast. Pipeline B provides 92% accuracy with moderate speed. Pipeline C provides the highest accuracy of 95%, but it is slow. This creates a direct trade-off between detection accuracy and processing speed.

Pipeline A is the most appropriate when very fast processing is required. It can respond quickly and is useful for applications where low latency is more important than maximum accuracy. However, its 88% accuracy means that it may miss or incorrectly classify some visual information compared with the other pipelines.

Pipeline B provides a stronger balance. Its 92% accuracy is higher than Pipeline A, while its moderate speed is still more practical than the slow speed of Pipeline C. This makes it suitable for many real-time or near-real-time computer vision applications.

Pipeline C gives the best accuracy at 95%, but its slow processing can cause delays. It is therefore more suitable for offline processing, detailed analysis, or applications where accuracy is more important than immediate response.

Conclusion

Pipeline B is the best balanced choice because it improves accuracy to 92% while maintaining moderate speed. Pipeline A should be used when strict speed requirements dominate, while Pipeline C should be used for accuracy-critical offline applications. The supplied assessment gives these exact accuracy/speed values. □filecite□turn1file0□L207-L214□

Question 23 – Multi-Model Comparison

The three models differ in accuracy, speed, and robustness. Model A has 88% accuracy, fast speed, and low robustness. Model B has 92% accuracy, moderate speed, and high robustness. Model C has 95% accuracy, slow speed, and high robustness.

Model A is attractive for applications where speed is the main requirement. However, its low robustness means its performance may decrease when the operating conditions change. Therefore, its fast response comes with a reliability trade-off.

Model B provides a balanced performance. It improves accuracy to 92%, has high robustness, and still maintains moderate speed. High robustness makes it more dependable across different conditions, while moderate speed makes it more practical than the slow Model C.

Model C gives the highest accuracy and high robustness, so it is the strongest choice when maximum detection quality is the priority. Its main disadvantage is slow processing, which can make it unsuitable for strict real-time applications.

Considering all three factors together, Model B provides the best overall balance. It does not have the highest accuracy or fastest speed, but it combines good accuracy, high robustness, and acceptable processing speed.

Conclusion

Model B is the best overall practical model. Model C is preferable for accuracy-critical applications, while Model A is preferable when speed is the dominant requirement. The assessment data for this comparison are 88/92/95% accuracy with Fast/Moderate/Slow speed and Low/High/High robustness.
□filecite□turn1file0□L181-L187□

Question 28 – Motion Model Evaluation

The two motion models have different strengths. Model A has high realism but low speed, while Model B has moderate realism and high speed. This creates a trade-off between visual/physical realism and computational performance.

Model A is able to generate more realistic motion. However, its low speed indicates higher computational requirements. Such a model may not respond quickly enough for interactive applications where frames must be generated continuously with low delay.

Model B has moderate realism but high speed. It can therefore be used when responsiveness is more important than maximum realism. This is useful for real-time applications such as interactive animation, games, or systems where immediate visual response is required.

For cinematic animation, where rendering can be performed offline and realism is a major objective, Model A is more appropriate. For real-time animation, Model B is the better choice because its high speed supports responsive operation.

Conclusion

The correct model depends on the application context: choose Model B for real-time animation where speed is important, and choose Model A for cinematic/offline animation where high realism is the priority.
□filecite□turn1file0□L215-L220□

Question 2 – Robust Detection under Environmental Variations

The given table compares Viola-Jones, YOLO, and Deep Learning under normal conditions, occlusion, and low-light conditions. Viola-Jones records 80%, 60%, and 55% respectively. YOLO records 90%, 75%, and 70%. Deep Learning records 95%, 85%, and 80%.

Under normal conditions, Deep Learning performs best with 95%, followed by YOLO with 90% and Viola-Jones with 80%. Under occlusion, Deep Learning again performs best at 85%, while YOLO gives 75% and Viola-Jones gives 60%. In low light, Deep Learning remains highest at 80%, followed by YOLO at 70% and Viola-Jones at 55%.

The average accuracy across the three conditions is approximately 86.7% for Deep Learning, 78.3% for YOLO, and 65.0% for Viola-Jones. This shows that Deep Learning provides the strongest overall performance. It is also more robust under difficult conditions such as occlusion and low light.

The trade-off is computational cost. Deep Learning provides the highest accuracy and robustness but can require greater computational resources. YOLO provides a useful middle ground because it offers good accuracy and is generally more suitable when speed is important. Viola-Jones can be useful where simplicity and speed are prioritized, but its accuracy falls considerably in difficult conditions.

Conclusion

Deep Learning is the best technique for overall robustness because it achieves the highest accuracy in normal, occluded, and low-light conditions. YOLO is the best compromise when computational or speed constraints are important. The source table provides the three-condition accuracy comparison.

□filecite□turn1file0□L20-L36□

Question 1 – Real-Time System Design Trade-off

The surveillance system has three choices. Viola-Jones is fast but has 78% accuracy. YOLO has moderate speed and 92% accuracy. Deep Learning has the highest accuracy at 95%, but it is slow. Therefore, the central issue is selecting a method that satisfies strict latency requirements without reducing accuracy too much.

Viola-Jones is the fastest option and can satisfy very strict latency constraints. However, its 78% accuracy is significantly lower than the other two methods. For a surveillance system, this could lead to more missed detections or incorrect detections.

YOLO provides a strong balance. Its 92% accuracy is much higher than Viola-Jones, while its moderate speed is more practical for real-time deployment than the slow Deep Learning approach. This makes YOLO the most suitable choice when the system requires both responsiveness and acceptable detection quality.

Deep Learning has the highest accuracy at 95%, but the source describes it as slow. Therefore, it is more appropriate when accuracy is the main objective and sufficient computing resources and processing time are available. It may not satisfy strict latency requirements.

Conclusion

YOLO should be selected for a practical real-time surveillance system because it provides 92% accuracy with moderate speed. Viola-Jones is suitable only when extremely low latency is more

important than accuracy, while Deep Learning is suitable when maximum accuracy is more important than latency. □filecite□turn1file0□L12-L19□

Question 7 – Stereo Vision vs Motion Estimation

Stereo Vision and Motion Estimation provide two different approaches for depth estimation. Stereo Vision is accurate but requires two cameras. Motion Estimation is less accurate but works with a single camera.

Stereo Vision uses information from two camera views to estimate depth. Its major advantage is higher accuracy. However, requiring dual cameras increases hardware requirements, cost, calibration requirements, and system complexity.

Motion Estimation can operate with a single camera, which makes it more practical for systems with limited hardware. The disadvantage is lower depth accuracy compared with Stereo Vision. Therefore, a single-camera system may have to accept reduced depth precision.

Under strict hardware constraints, Motion Estimation is preferable because it can achieve depth estimation without requiring a second camera. When hardware constraints are relaxed and accurate depth is the priority, Stereo Vision is the better option.

Conclusion

Motion Estimation is the preferred approach when hardware is limited because it requires only one camera. Stereo Vision should be selected when accurate depth is more important and two cameras can be supported. □filecite□turn1file0□L62-L67□

Question 11 – Detection Model Comparison under Resource Constraints

The three models have the following performance: Model A has 85% accuracy and 40 ms latency, Model B has 92% accuracy and 90 ms latency, and Model C has 90% accuracy and 60 ms latency.

Model A has the lowest latency, making it the fastest option for real-time systems. Its 85% accuracy is lower than Models B and C, but it may still be acceptable when the application has moderate accuracy requirements.

Model B gives the highest accuracy at 92%, but its 90 ms latency is the highest among the three. Therefore, it is less suitable when immediate response is critical. It can be selected when improved accuracy is more important than latency.

Model C provides 90% accuracy with 60 ms latency. It is a useful middle option because it improves accuracy over Model A while keeping latency considerably lower than Model B. If the application can tolerate 60 ms latency, Model C provides a good accuracy-speed balance.

For strict real-time systems with moderate accuracy requirements, Model A is the optimal choice because its 40 ms latency is the lowest. If the system can accept additional latency for better accuracy, Model C is the stronger balanced alternative.

Conclusion

Model A is optimal under strict latency constraints because it has the lowest latency of 40 ms and still provides 85% accuracy. Model C is the best alternative when a moderate increase in latency is

acceptable. The supplied data give the exact accuracy and latency values. [filecite turn1file0 L94-L101](#)

Overall Summary

Across the selected questions, the decisions depend on balancing accuracy, robustness, speed, hardware requirements, and computational cost. High-accuracy methods are not always the best for real-time systems because increased accuracy may require additional computation. Similarly, very fast methods may sacrifice accuracy or robustness. Therefore, the most appropriate computer-vision technique should always be selected according to the actual application requirements.

This approach follows the assessment rubric, which evaluates understanding of the data, analysis and justification, application of concepts, conclusions, and clarity. [filecite turn1file0 L242-L365](#)